Semester	Course Code	Course Category	Hours/ Week	Credits	Marks for Evaluation		
					CIA	ESE	Total
I	23MCA1CC4	CORE – IV	4	3	25	75	100

Course Title	Resource Management Techniques

SYLLABUS		
Unit	**Contents**	**Hours**
I	Operations Research – Nature and Features of O.R. – Definitions of O.R. – Applications of O.R. – Linear Programming Problem – Mathematical Formulation of the Problem – Graphical Solution Method – Some Exceptional Cases – Simplex Method – The Computational Procedure – Use of Artificial Variables – *Two-Phase Method* – Big-M Method.	12
II	Transportation Problem – *Linear Programming Formulation of the Transportation Problem* – Finding an Initial BFS – North-West Corner Rule – Matrix Minima Method – Vogel's Approximation Method – Test for Optimality – Assignment Problem – Mathematical Formulation of the problem – Hungarian Assignment Method – Special Cases in Assignment Problems – The Travelling Salesman Problem.	12
III	Network Scheduling by PERT / CPM – Network: Basic Components – Logical Sequencing – Rules for Network Construction – Concurrent Activities – Critical Path Analysis – Probability Considerations in PERT – *Distinction between PERT and CPM* – Applications of Network Techniques.	12
IV	Inventory Control – Types of Inventories – Reasons for Carrying Inventories – The Inventory Decisions – Objectives of Scientific Inventory Control – Costs Associated with Inventories – *Factors Affecting Inventory Control* – An Inventory Control Problem – The Concept of EOQ – Deterministic Inventory Problems with No Shortages – Deterministic Inventory Problems with Shortages – ABC Analysis (Always, Better, Control) Technique.	12
V	Queueing Theory – Queueing System – Elements of a Queuing System – Operating Characteristics of a Queueing System – Classification of Queuing Models – *Definition of Transient and Steady States* – Queueing Models – (M/M/1):(∞ /FIFO) – (M/M/1):(N/FIFO) – (M/M/C):(∞/FIFO) – (M/M/C):(N/FIFO). .	12
VI	**Current Trends (For CIA only):** Recent Trends in Optimization Techniques	

...... Self Study

Text Book:
Kanti Swarup, P.K. Gupta and Man Mohan, *Operations Research*, Sultan Chand & Sons Educational Publishers, New Delhi, Sixteenth Edition, Reprint 2013.
Reference Book(s):
1. Hamdy A. Taha, *Operations Research : An Introduction*, PHI, New Delhi, 8th Edition, 2008. 2. A. Ravindran, Don T. Phillips, James J. Solberg, *Operations Research Principles and Practice*, John Wiley & Sons, Second Edition, Third Reprint, 2007.
Web Resource(s):
1. https://www.tutorialsduniya.com/notes/operational-research-notes/ 2.https://www.researchgate.net/publication/313880623_Introduction_to_Operations_Research_Theory_and_Applications 3. https://archive.nptel.ac.in/courses/111/105/111105039/ 4. https://archive.nptel.ac.in/courses/112/106/112106134/

Course Outcomes		
Upon successful completion of this course, the student will be able to:		
CO No.	**CO Statement**	**Cognitive Level (K-Level)**
CO1	Recall the fundamental concepts involved in various Optimization Methods	K1
CO2	Summarize the procedure for solving different Operations Research Problems	K2
CO3	Apply the concept of selected resource management techniques to solve the real-life problems	K3
CO4	Analyze and Examine the steps involved in decision-making problems in management	K4
CO5	Design, Develop and Explain the suitable optimization technique and then solve the real-world scientific and business problems	K5 & K6

Relationship Matrix:

Course Outcomes (COs)	Programme Outcomes (POs)					Programme Specific Outcomes (PSOs)					Mean Score of COs
	PO1	**PO2**	**PO3**	**PO4**	**PO5**	**PSO1**	**PSO2**	**PSO3**	**PSO4**	**PSO5**	
CO1	3	3	1	1	3	3	2	2	2	3	2.3
CO2	3	3	3	3	2	3	3	3	2	0	2.5
CO3	3	3	3	3	3	2	3	3	2	2	2.7
CO4	2	3	2	2	3	3	3	1	2	2	2.3
CO5	3	3	3	3	2	2	3	3	3	3	2.8
										Mean Overall Score	2.52
										Correlation	High

Mean Overall Score = Sum of Mean Score of COs / Total Number of COs

Mean Overall Score	**Correlation**
< 1.5	Low
≥ 1.5 and < 2.5	Medium
≥ 2.5	High

Course Coordinator: Dr. O.A. Mohamed Jafar